



4Arms – Colles Distal Forearm Fracture Reduction Model

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Problem Statement

Hospitals lack effective tools to train residents on treating one of the most common wrist fractures, a distal radius fracture (Figures 1 & 2). Problems with existing models are that they are expensive, fragile, and inaccurate, discouraging hospital investment.

The team aims to develop a realistic, durable, and safe model for training closed reductions at the Duke Hospital.



Figure 1: X-Ray of a healthy wrist.



Figure 2: X-Ray of a fractured distal radius in the wrist.

Existing Prototype

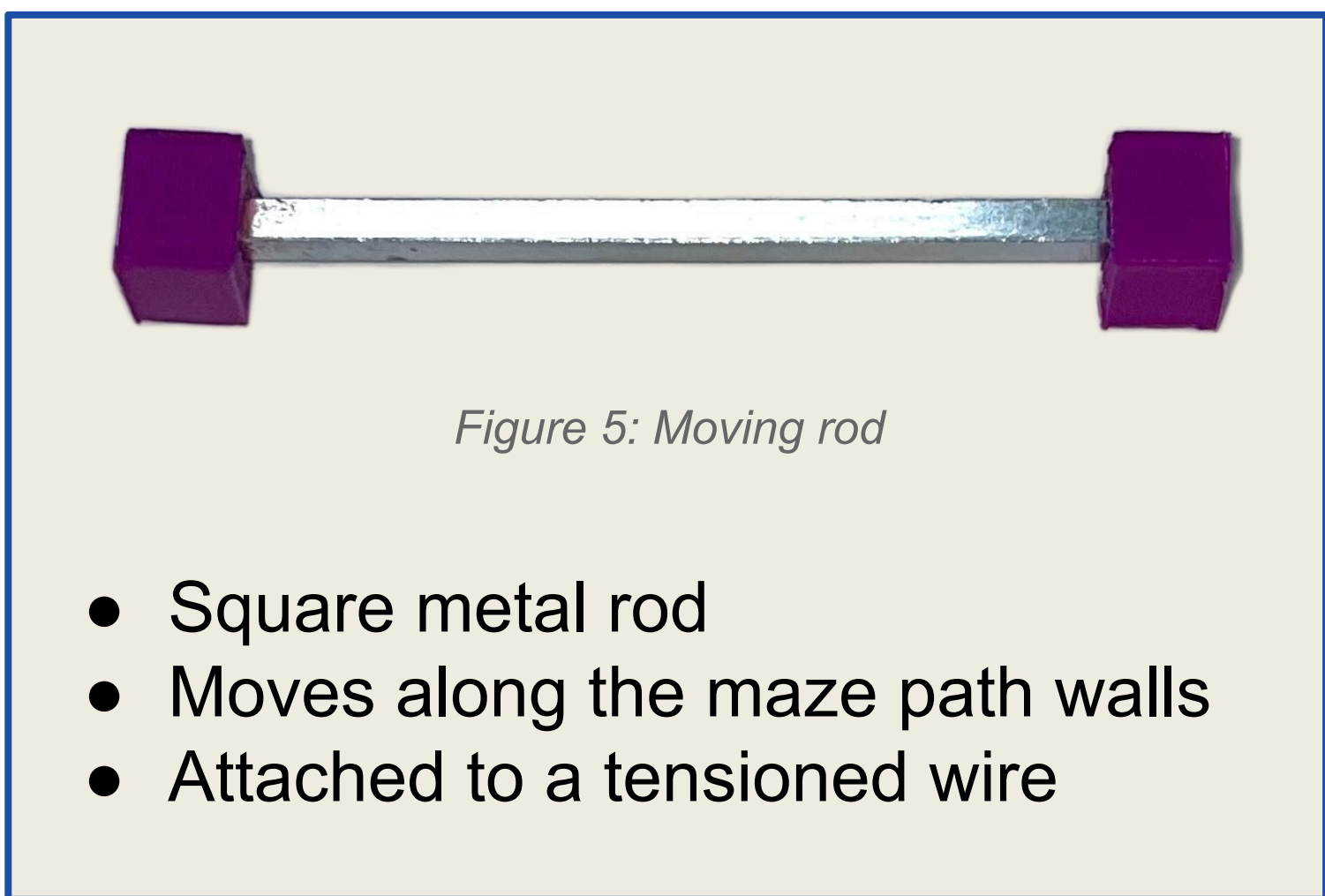


Figure 3: Existing prototype created by engineer David Mercer

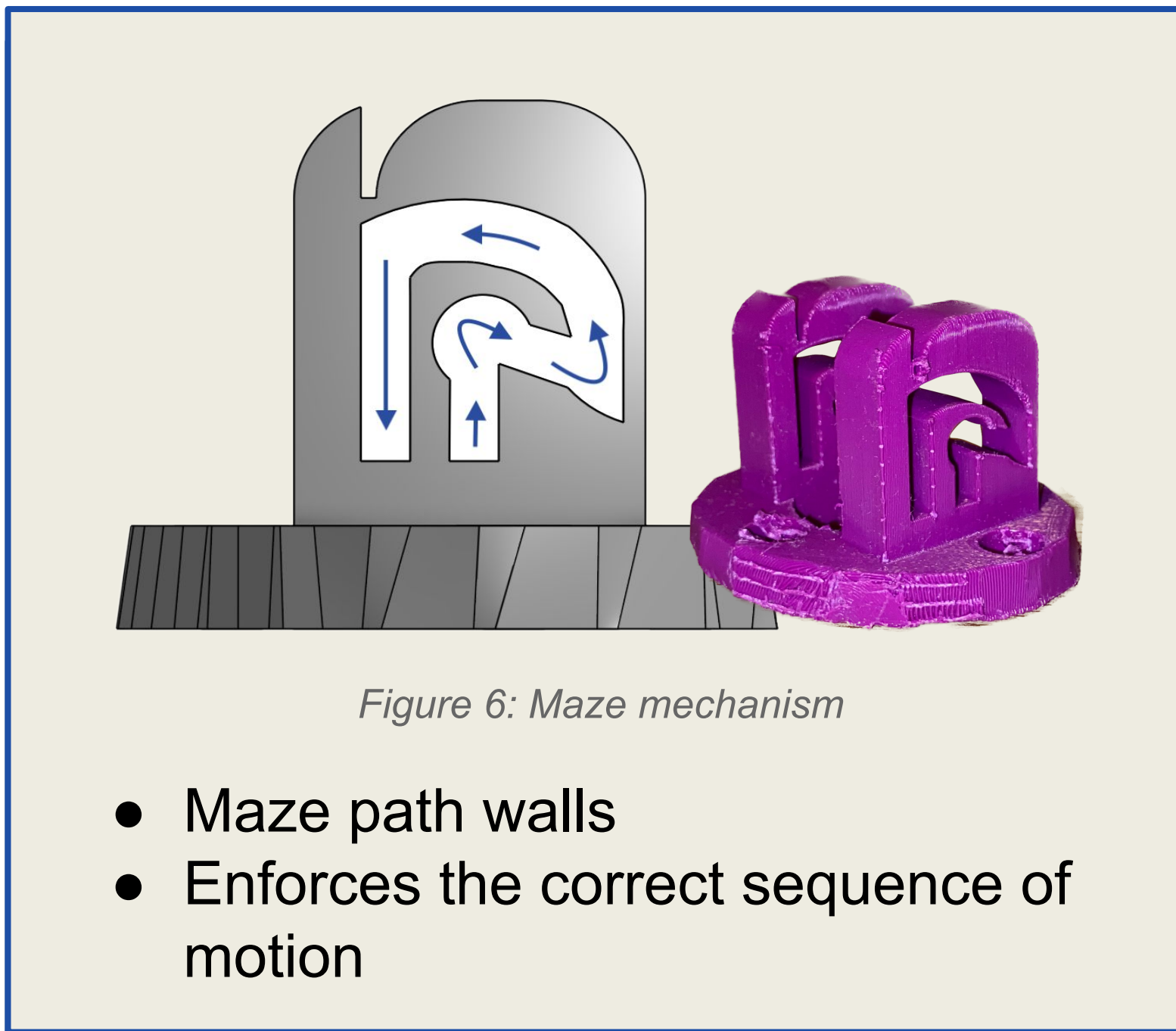
An existing prototype features a 3D-printed hand and forearm, a spring to replicate forces, and a "step" mechanism to mimic the fracture. Issues to address include:

- Unrealistic horizontal movement
- Does not enforce correct sequence of motion
- Overall lack of realism

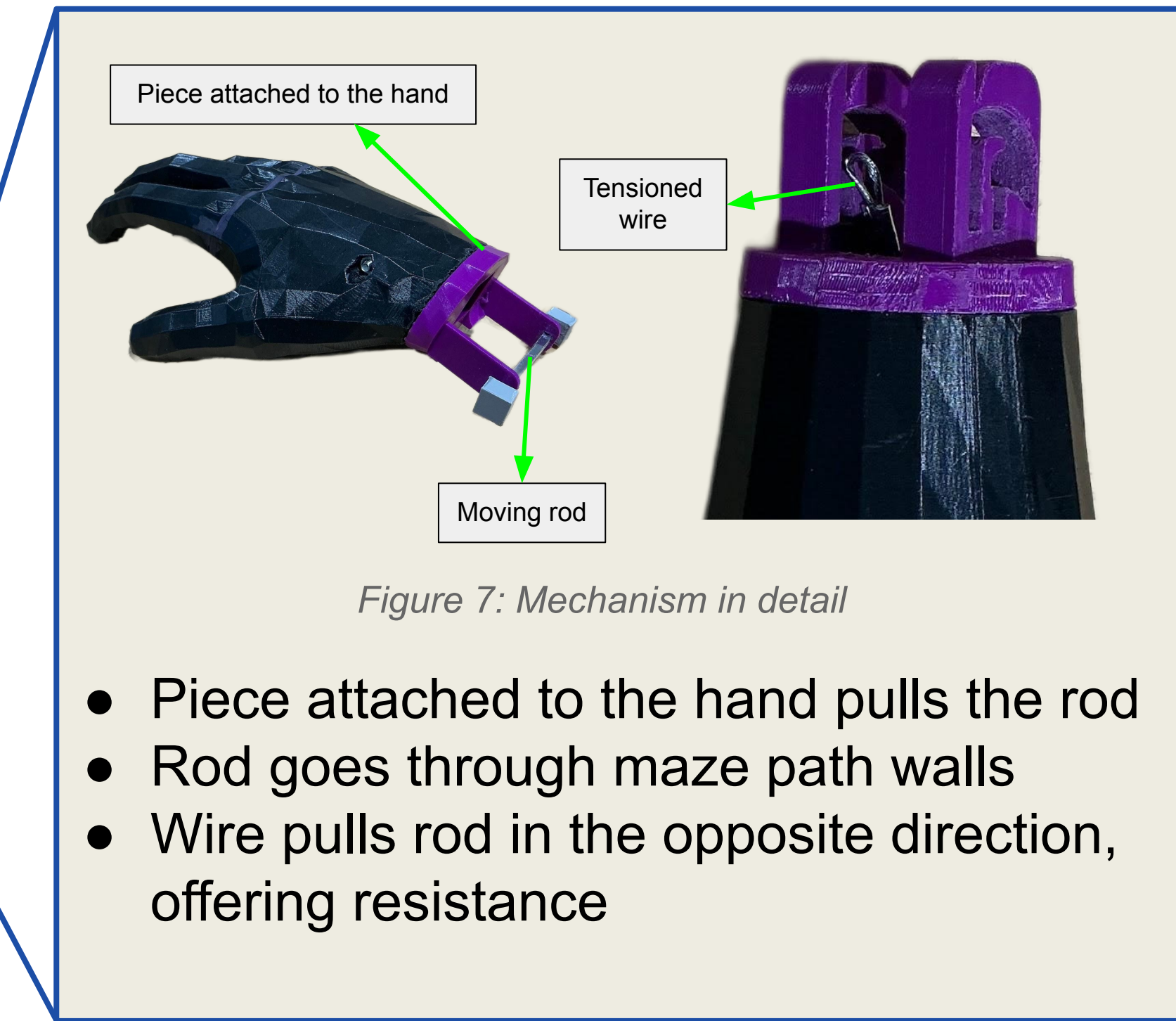
Final Prototype Model



- Square metal rod
- Moves along the maze path walls
- Attached to a tensioned wire



- Maze path walls
- Enforces the correct sequence of motion



- Piece attached to the hand pulls the rod
- Rod goes through maze path walls
- Wire pulls rod in the opposite direction, offering resistance



- Removable neoprene sleeve
- Improves realism with skin-like feel
- Covers fracture zone

Figure 4: Final Prototype with visible attached mechanism

Design Criteria and Testing

Design Criteria	Procedure	Target Value	Results
Accurate Mechanism & Forces	8 physicians performed 10 reductions each using the arm and were asked to rate the accuracy of the force and of the reduction movements on a 10-point Likert Scale.	$\geq 8.0/10$ Average Motion $\geq 8.0/10$ Average Force	Motion: Pass (8.3) Force: Pass (8.1)
Safety	20 students used the device and reported on the safety perceptions on a 5-point Likert scale.	$\geq 4.5/5$ Average	Pass (4.9)
Durability	The full process of reduction and reset were repeated by team members, students, and physicians to assess the frequency or ease of damage on a 3-point scale.	$\geq 2.8/3$ Average	Pass (2.8)
Easy Reset	20 undergraduate students were asked to reset the reduced arm while being timed, in addition to answering a user-defined scale.	< 5 seconds $\geq 2.5/3$ Average	Fail (10.3 seconds) Pass (2.5)
Realistic	35 blindfolded people evaluated the look and feel of the model and rated the material's realism on a 5-point Likert scale.	$\geq 3.5/5$ Average	Fail (3.1)

Conclusion & Future Work

The team's design was successful, achieving its main objective of accurately replicating a distal forearm reduction maneuver, in addition to the approval of the mechanism and forces by physicians. However, there remains room for improvement. Moving forward the group aims to:

- Design an automated mechanism for force calibration and adjustment.
- Make the model more realistic with materials like silicone.
- Increase the durability by creating a titanium maze (rather than plastic).



Figure 9: Final prototype being used by resident at the Duke Hospital.

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<https://radiopaedia.org/cases/normal-wrist-x-rays>
<https://www.orthobullets.com/trauma/1027/distal-radius-fractures>